
Case Study 4:

Model Predictive Control for Wind Power Smoothing with Controlled Battery Storage

References

● This case study is based on the following published research work:

[1] M. Khalid and A. V. Savkin, "Development of Short-Term Prediction System for Wind Power Generation Based on Multiple Observation Points," in *Proceedings of the International Conference on Sustainability in Energy and Buildings*, Brighton, UK, Apr 2009.

[2] M. Khalid and A. V. Savkin, "Model Predictive Control for Wind Power Generation Smoothing with Controlled Battery Storage," in *Proceedings of the 48th IEEE Conference on Decision and Control*, Shanghai, China, pp. 7849-7853, Dec 2009.

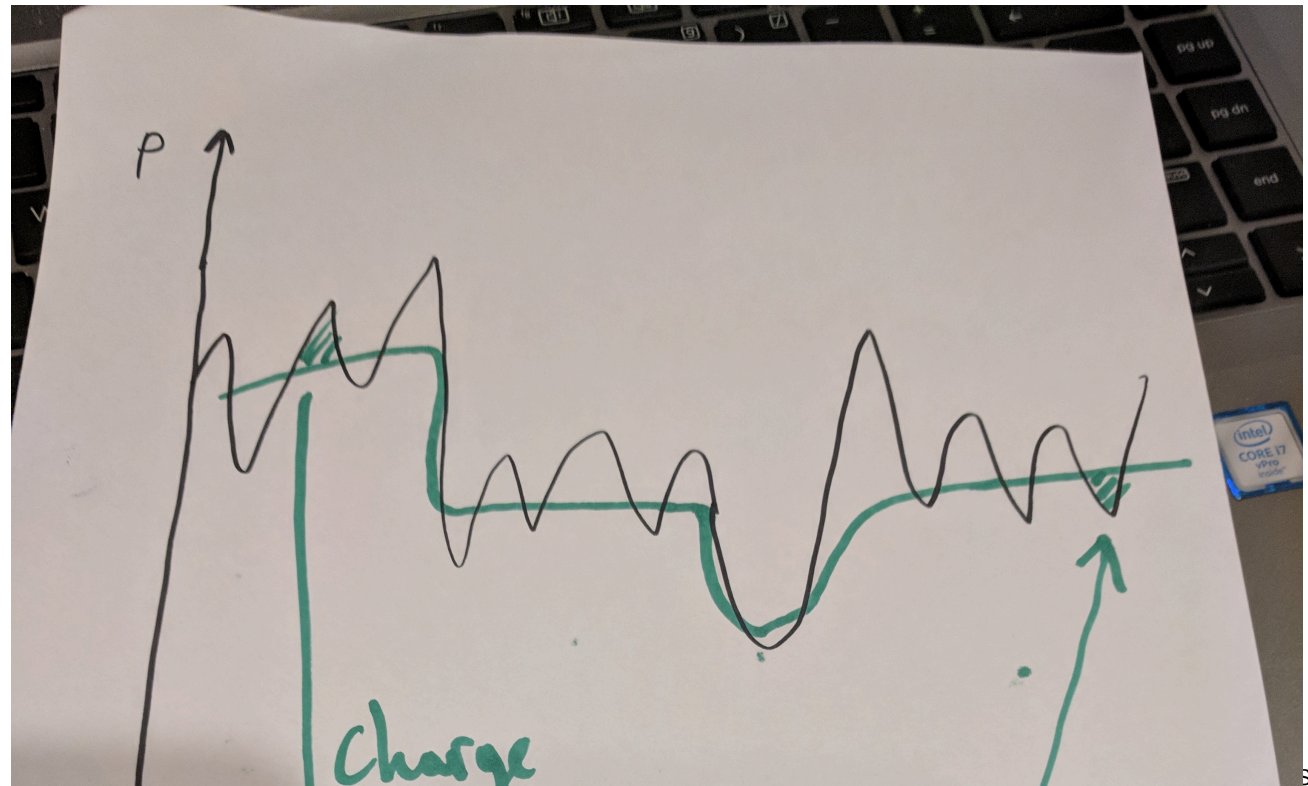
Introduction

- Wind energy is in highest demand because it is,
 - renewable
 - clean



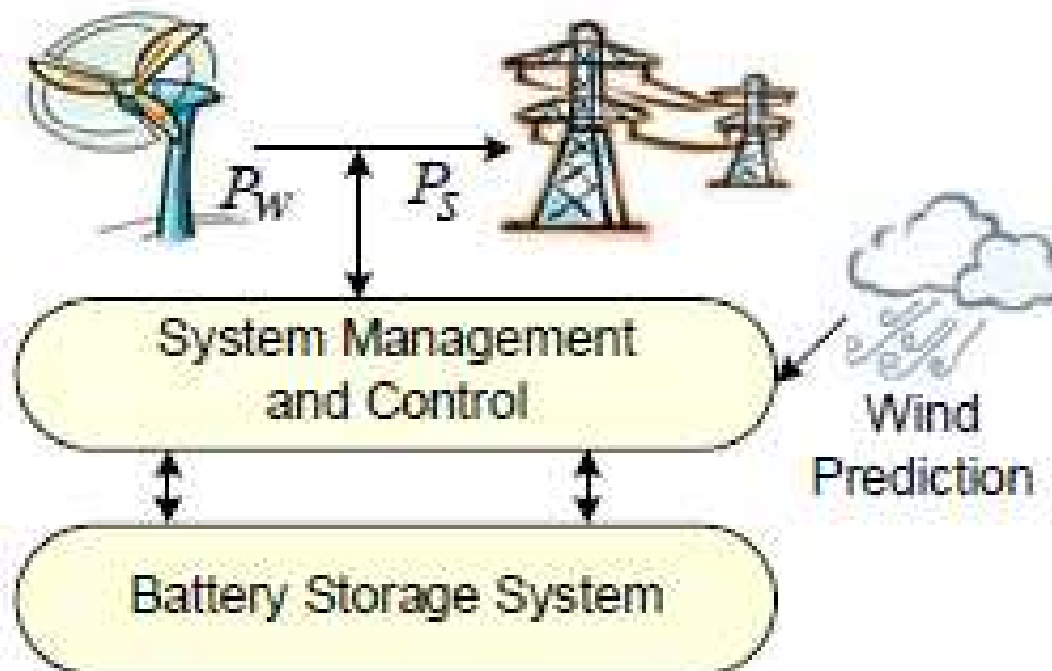
Introduction (Contd...)

- Now a days, wind energy can be considered as a valuable supplement to conventional energy sources but,
 - the drawback is that wind is a highly fluctuating resource,
 - the efficient and cost effective integration of wind power into electricity grid has become the greatest challenge.



Introduction (Contd...)

- An efficient control system is required in order to absorb a large fraction of wind power in the electrical systems, which should be capable of,
 - properly smoothing the wind farm power output,
 - optimizing the wind energy storage in batteries.



Objectives

- To smooth the wind farm power output based on battery storage.
- To develop a controller based on model predictive control (MPC) to optimize the wind energy storage in batteries.
- To propose a wind prediction system to facilitate and improve the overall control system.

Motivation

- The development of an efficient control system overcomes significant problems associated with wind energy e.g.
 - the frequency control,
 - the voltage support,
 - the excessive peak loads on transmission lines,
 - the increasing demand for high ramp rate backup systems etc.
- Optimizing the wind energy storage in batteries,
 - enhances the battery life and efficiency,
 - reduces the associated cost.
- Therefore, an efficient control system makes it possible to absorb a large fraction of wind power in the electrical systems.

Wind Farm Data

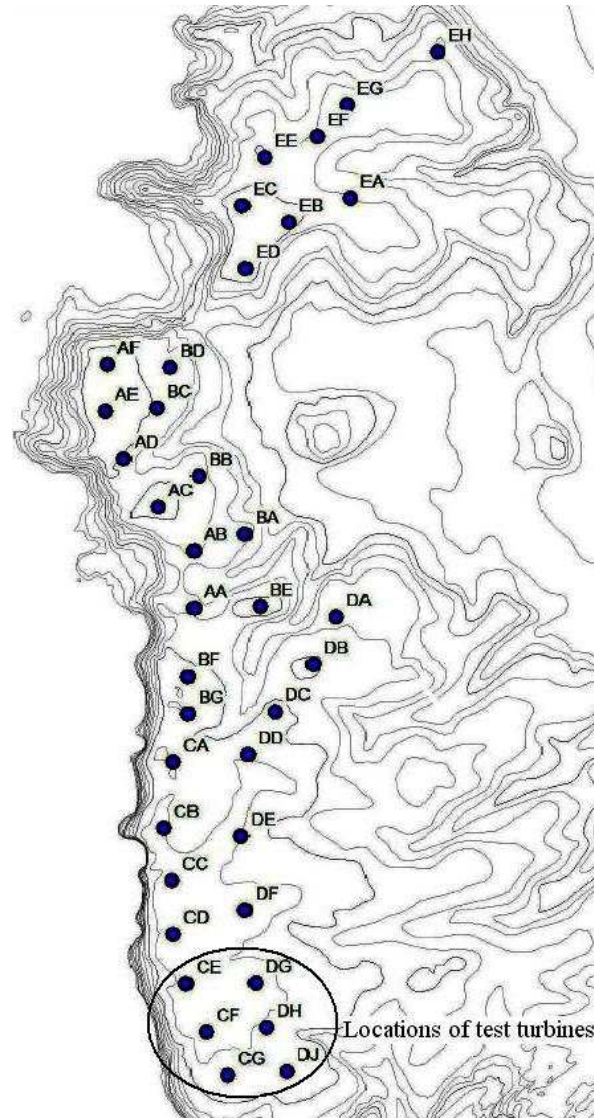
- The real world data (wind speed, direction and power) used in this study is obtained from Woolnorth wind farm in Tasmania, Australia as shown,



- The wind farm observation data has 10 min resolution measured at hub height at each of the 37 turbines.
- The individual locations of turbines and the turbines under test are highlighted in the following figure,

Data (Contd...)

- The location of test turbines in Woolnorth wind farm,



Data (Contd...)

- The Woolnorth wind farm at a cliff,



The Control System Model

- We propose the following simple linear dynamic model to smooth the wind power variations and to optimize the wind energy storage in batteries:

$$\begin{aligned}x_1(k+1) &= r(k) - u(k) \\x_2(k+1) &= x_1(k) + x_2(k)\end{aligned}\tag{1}$$

$$y(k) = x_2(k)\tag{2}$$

with the following cost function,

$$J := \sum_{k=0}^{M-1} (r(k) - u(k))^2 + \alpha(x_2(k) - L)^2 \rightarrow \min\tag{3}$$

The Model (Contd...)

where

- $r(k)$ is the real wind power (P_w),
- $u(k)$ is the smoothed power (P_s),
- $x_1(k)$ is the error between real and smoothed power,
- $x_2(k)$ is the battery load,
- L is the desired storage state of the battery,
- M is the control horizon,
- the parameter α is the weighting factor associated with the storage state control,

The Model (Contd...)

subject to following constraints,

$$0 \leq u(k) \leq c_1; k = 0, 1, \dots, M - 1 \quad (4)$$

$$-c_2 \leq u(k + 1) - u(k) \leq c_2; k = 0, 1, \dots, M - 1 \quad (5)$$

$$0 \leq x_2(k) \leq c_3; k = 0, 1, \dots, M - 1 \quad (6)$$

The Model (Contd...)

where

- c_1 , c_2 and c_3 are the constraints,
- The cost function (3) penalizes the deviations of the wind smoothed power from the actual wind power, and the deviations of the state of the battery and the desired state L .
- The constraint (4) keeps the smoothed power between 0 and rated value of the turbine.
- The constraint (5) is the difference between two consecutive values of the smoothed power, it controls the degree of smoothing.
- The constraint (6) keeps the storage under battery capacity and prevents the negative load on the battery.

The Controller Setup

- The MPC is based on the solution of an on-line optimal control problem where,
 - a receding horizon approach is utilized in such a way that for any current state $x(k)$ at time k , an open loop optimal control problem is solved over some future interval taking into account the current and future constraints.
- The MPC algorithm calculates an open loop sequence of the manipulated variables in such a way that the future behavior of the plant is optimized where,
 - the first value from this optimal sequence is injected into the plant.
- The procedure is repeated at time $(k + 1)$ using the current state $x(k + 1)$.

The Controller (Contd...)

- The MPC is based on the the optimal solution of the form,

$$U^{Opt} = [\bar{u}^{OPT}(0), \bar{u}^{OPT}(1), \dots, \bar{u}^{OPT}(M - 1)]^T \quad (7)$$

which can be numerically solved as static-optimization problem of the form,

$$U^{OPT} = \arg \min_U U^T W U + 2U^T V \quad (8)$$

$LU \leq K$

with constraints expressed in the form,

$$LU \leq K \quad (9)$$

using [Standard Quadratic Programming](#) algorithms.

The Controller (Contd...)

- We are converting our linear constraints and quadratic cost function associated with the model to the standard form, and following matrices are formulated,

$$W = \alpha(\bar{C}\psi)^T \bar{C}\psi + I \quad (10)$$

$$V = \alpha[(\bar{C}\phi)^T (\bar{C}\psi)]^T - \alpha(\bar{C}\psi)^T L - R \quad (11)$$

where

$$\phi = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^M \end{bmatrix} + \begin{bmatrix} B_2 & 0 & \dots & 0 & 0 \\ AB_2 & B_2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A^M B_2 & A^{M-1} B_2 & \dots & \dots & B_2 \end{bmatrix} R \quad (12)$$

The Controller (Contd...)

$$\psi = \begin{bmatrix} B_1 & 0 & \cdots & 0 & 0 \\ AB_1 & B_1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A^M B_1 & A^{M-1} B_1 & \cdots & \cdots & B_1 \end{bmatrix} \quad (13)$$

$$\bar{C} = \begin{bmatrix} C & 0 & \cdots & 0 \\ 0 & C & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & C \end{bmatrix}, R = \begin{bmatrix} r(0) \\ \hat{r}(1) \\ \vdots \\ \hat{r}^{M-1} \end{bmatrix} \quad (14)$$

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, B_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, B_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (15)$$

The Controller (Contd...)

$$L = \begin{bmatrix} L & \cdots & L \end{bmatrix}^T, C = \begin{bmatrix} 0 & 1 \end{bmatrix}^T \quad (16)$$

● The constraint matrices are expressed as,

$$L = \begin{bmatrix} I & -I & J & -J & D & -D \end{bmatrix}^T \quad (17)$$

where

● I is $(M \times M)$ identity matrix and J is following $(M \times M)$ matrix,

$$J = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \quad (18)$$

The Controller (Contd...)

and

$$D = \begin{bmatrix} CB_1 & 0 & \dots & 0 & 0 \\ CAB_1 & CB_1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^M B_1 & CA^{M-1} B_1 & \dots & \dots & CB_1 \end{bmatrix} \quad (19)$$

$$K = \begin{bmatrix} \bar{C}_{1 \max} & \bar{C}_{1 \min} & \bar{C}_{2 \max} & \bar{C}_{2 \min} & \bar{C}_{3 \max} & \bar{C}_{3 \min} \end{bmatrix}^T \quad (20)$$

where

The Controller (Contd...)

$$\bar{C}_{1\max} = \begin{bmatrix} c_1 & \cdots & c_1 \end{bmatrix}^T \quad (21)$$

$$\bar{C}_{1\min} = \bar{C}_{3\min} = \begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}^T \quad (22)$$

$$\bar{C}_{2\max} = \begin{bmatrix} u(0) + c_2 & c_2 & \cdots & c_2 \end{bmatrix}^T \quad (23)$$

$$\bar{C}_{2\min} = \begin{bmatrix} -u(0) - c_2 & -c_2 & \cdots & -c_2 \end{bmatrix}^T \quad (24)$$

The Controller (Contd...)

$$\bar{C}_{3 \max} = \begin{bmatrix} c_3 \\ c_3 \\ \vdots \\ c_3 \end{bmatrix} - \begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^{M-1} \end{bmatrix} - FR \quad (25)$$

with

$$F = \begin{bmatrix} CB_2 & 0 & \dots & 0 & 0 \\ CAB_2 & CB_2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^M B_2 & CA^{M-1} B_2 & \dots & \dots & CB_2 \end{bmatrix} \quad (26)$$

- The optimization problem at each step is solved using MATLAB.

The Controller Stability

- It is clear that the system (1), (2) with our model predictive controller is stable.
- Indeed, stability means that for any bounded input $r(k)$ of the system (1), (2), $x_1(k)$, $x_2(k)$, and $u(k)$ are bounded.
- The constraint (4) obviously implies that $u(k)$ is bounded, and the constraint (6) implies that $x_2(k)$ is bounded.
- Furthermore, the first equation in our proposed model (1) implies that $x_1(k)$ is bounded for bounded $r(k)$ and $u(k)$.

The Proposed Prediction Model

- We are proposing a wind power prediction model which is used along with MPC to improve the performance of overall control system.
- This model consists of two stages,
 - wind speed and direction prediction stage,
 - wind power prediction stage.

Stage 1: Wind Speed and Direction Prediction

- Generally this model can be represented as,

$$Y = MX \quad (27)$$

where

$$Y = [Y(t+1) \ \cdots \ Y(t-N+1)]^T \quad (28)$$

$$M = \begin{bmatrix} V_1 & \cdots & V_K \end{bmatrix} \quad (29)$$

$$X = [c_1 \ \cdots \ c_K]^T \quad (30)$$

Prediction Model (Contd...)

• The matrix $V_i \in \mathbb{R}^{n \times n_i}$ is,

$$V_i = \begin{bmatrix} \mathbf{v}_i(t)^T \\ \mathbf{v}_i(t-1)^T \\ \vdots \\ \mathbf{v}_i(t-N)^T \end{bmatrix} \quad (31)$$

with

$$\mathbf{v}_i(t) = \begin{bmatrix} v_i(t) & v_i(t-1) & \cdots & v_i(t-n_i) \end{bmatrix}^T. \quad (32)$$

Prediction Model (Contd...)

- The first element of the vector Y is,

$$Y(t+1) = \sum_{i=1}^K c_i^T \mathbf{v}_i(t) \quad (33)$$

with

$$c_i = [\alpha_{i,1} \ \cdots \ \alpha_{i,n_i}]^T \in \Re^{n_i} \quad (34)$$

Prediction Model (Contd...)

where

- K is the number of the turbines,
- n_i is the order of autoregressive (AR) models of the i -th turbine,
- N is the number of the historical data points taken,
- $v_i(\xi)$ is the wind speed recorded at the i -th turbine at time ξ ,
- The vector c_i contains the coefficients of the AR model of the i -th turbine.

Prediction Model (Contd...)

where

- The vector X containing the coefficients of the model is estimated using the method of adaptive least squares at each time step. The following relation is used to compute X ,

$$X = (M^T M)^{-1} M^T Y \quad (35)$$

- All the data from turbines is converted to wind vectors v_x and v_y as,

$$\begin{aligned} v_x(t) &= v(t) \cos \theta(t) \\ v_y(t) &= v(t) \sin \theta(t) \end{aligned} \quad (36)$$

where

- $\theta(t)$ is the direction of the wind in degrees measured at time t .

Prediction Model (Contd...)

- The proposed model in (27) is utilized to predict the individual wind vectors denoted as $\hat{v}_x$ and $\hat{v}_y$.
- Finally, the prediction of wind speed and direction at each time step is obtained using the following relation,

$$\hat{v}(t+1) = \sqrt{\hat{v}_x(t+1)^2 + \hat{v}_y(t+1)^2} \quad (37)$$

and

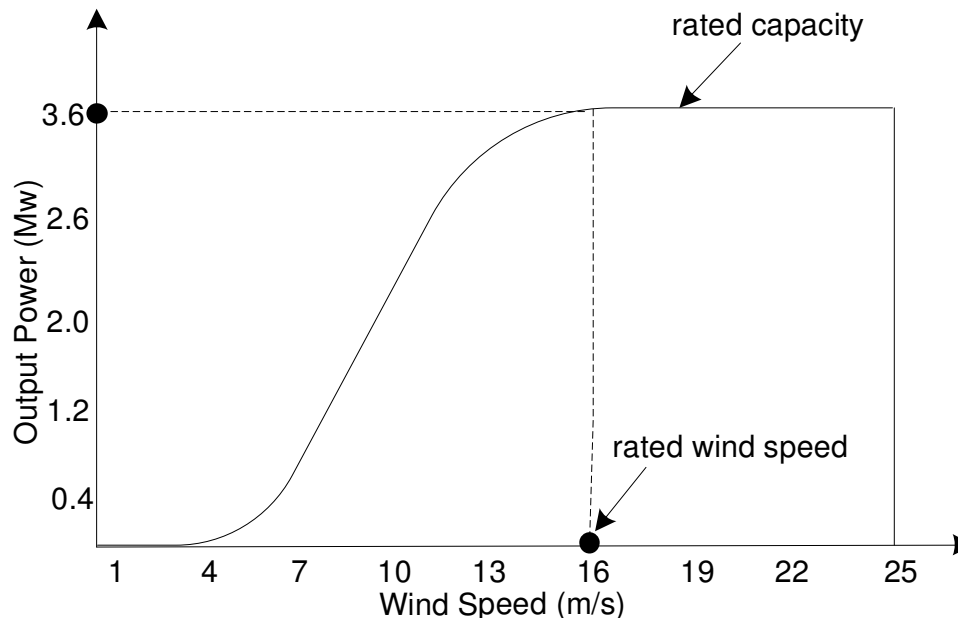
$$\hat{\theta}(t+1) = \tan^{-1}\left(\frac{\hat{v}_y(t+1)}{\hat{v}_x(t+1)}\right) \quad (38)$$

where

- $\hat{v}(t+1)$ and $\hat{\theta}(t+1)$ are the speed and direction prediction for $t+1$ made at t respectively.

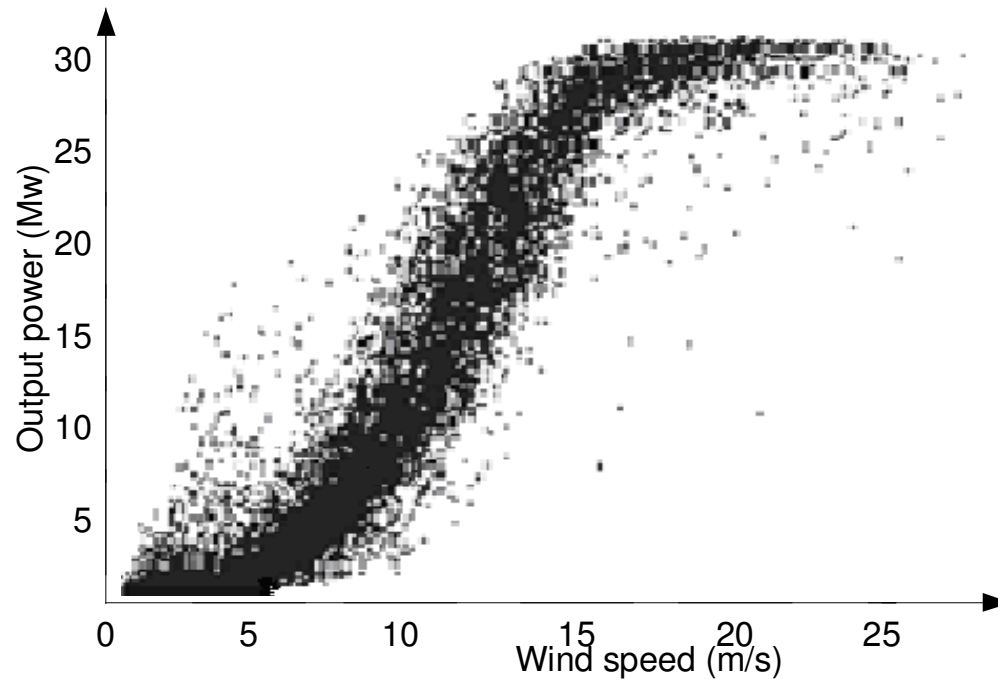
Stage 2: Wind Power Prediction

- Based on predicted wind speed and direction in stage 1, the predicted power output is achieved by feeding the predicted wind speed to the input of appropriate power curve (PC) model.
- The PC of a wind turbine is a graph that indicates how large the electrical power output will be for the turbine at different wind speeds as shown below:



Stage 2: Wind Power Prediction (Contd...)

- However, the deterministic machine power curve is different from the empirical power curve obtained in the real operation at a wind farm as shown below:



- That's why the power curve is derived from measured wind speed and power.

Stage 2: Wind Power Prediction (Contd...)

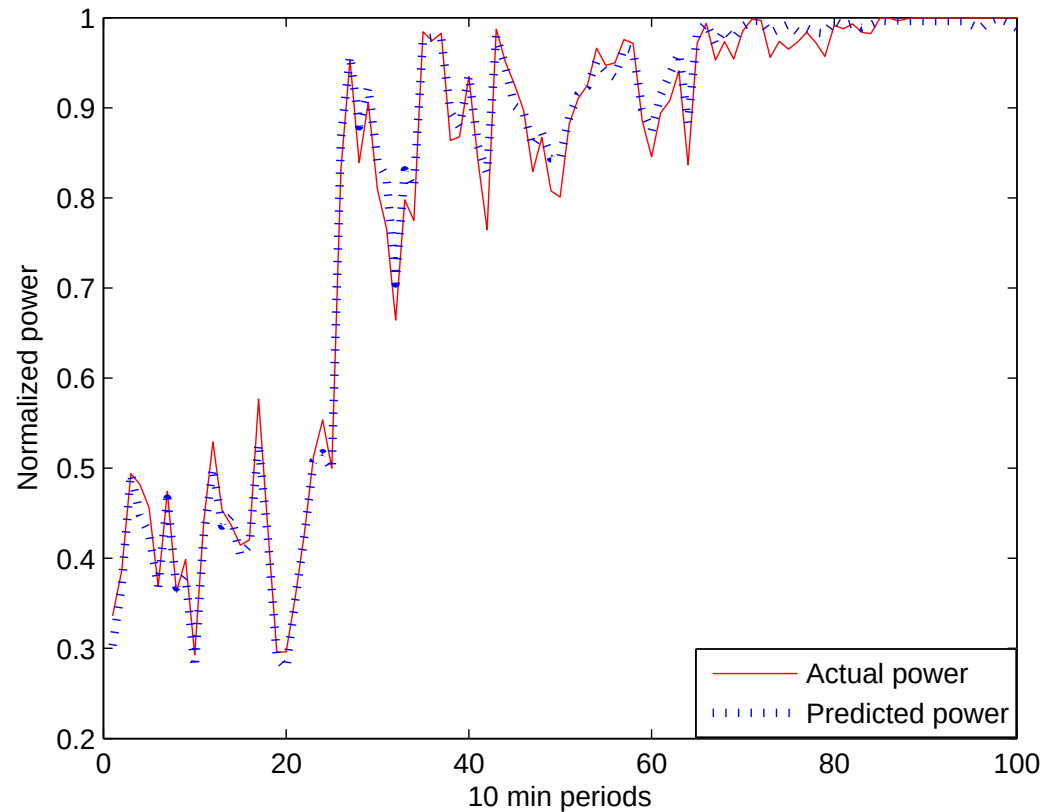
- The wind power generation is also influenced by wind direction. However, the influence of wind direction on power output is less significant as compared with wind speed due to the fact that turbines are directed to face the wind during its operation.
- It has been concluded by experiments that, for a given site, the direction dependent PC models can lead to the accuracy improvement in final wind power prediction.
- Therefore, our PCs are based on direction distribution of the wind at the given wind farm site.

Stage 2: Wind Power Prediction (Contd...)

- In our case, PC is direction dependent empirical power curve based on the historical data of the wind farm under study.
- Each direction is of 90° , hence 4 direction sectors are considered in total.
- The power curve for each chosen direction sector is modeled using the standard method of bins.
- Our prediction model is capable of multi-step ahead predictions as well, which are effectively used in the optimization side of MPC based controller.

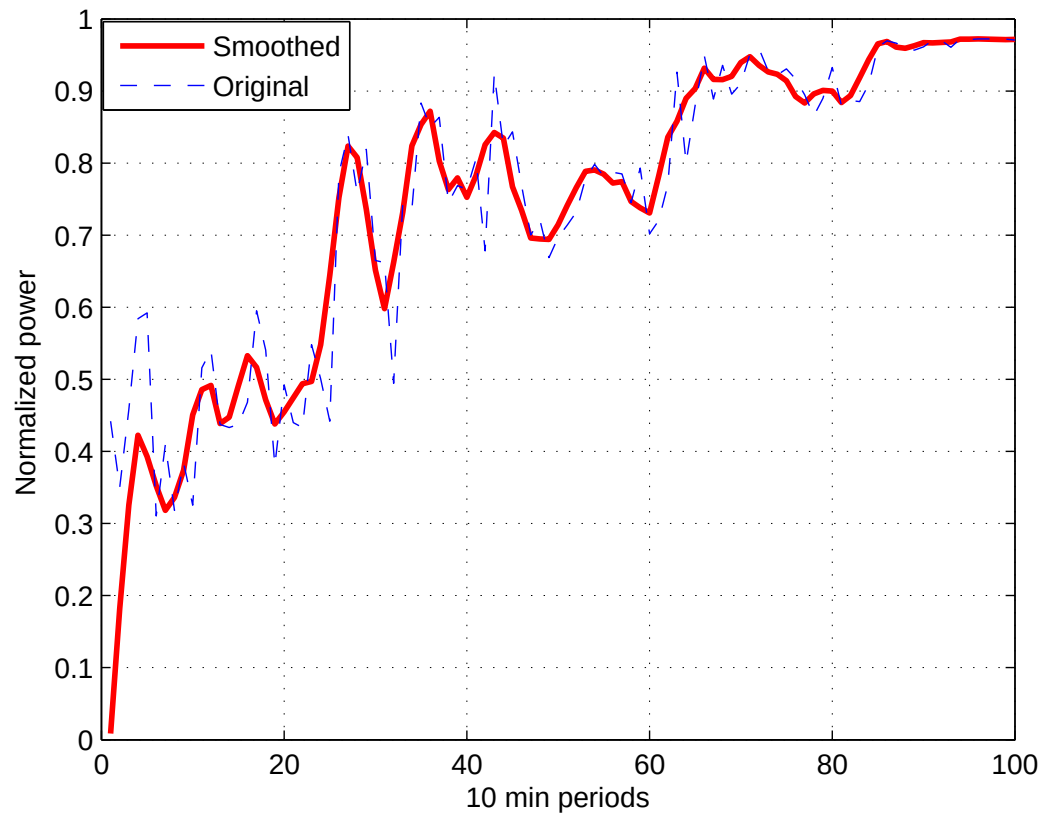
The Performance of Prediction Model

- The performance of our proposed prediction model is shown in the following figure,



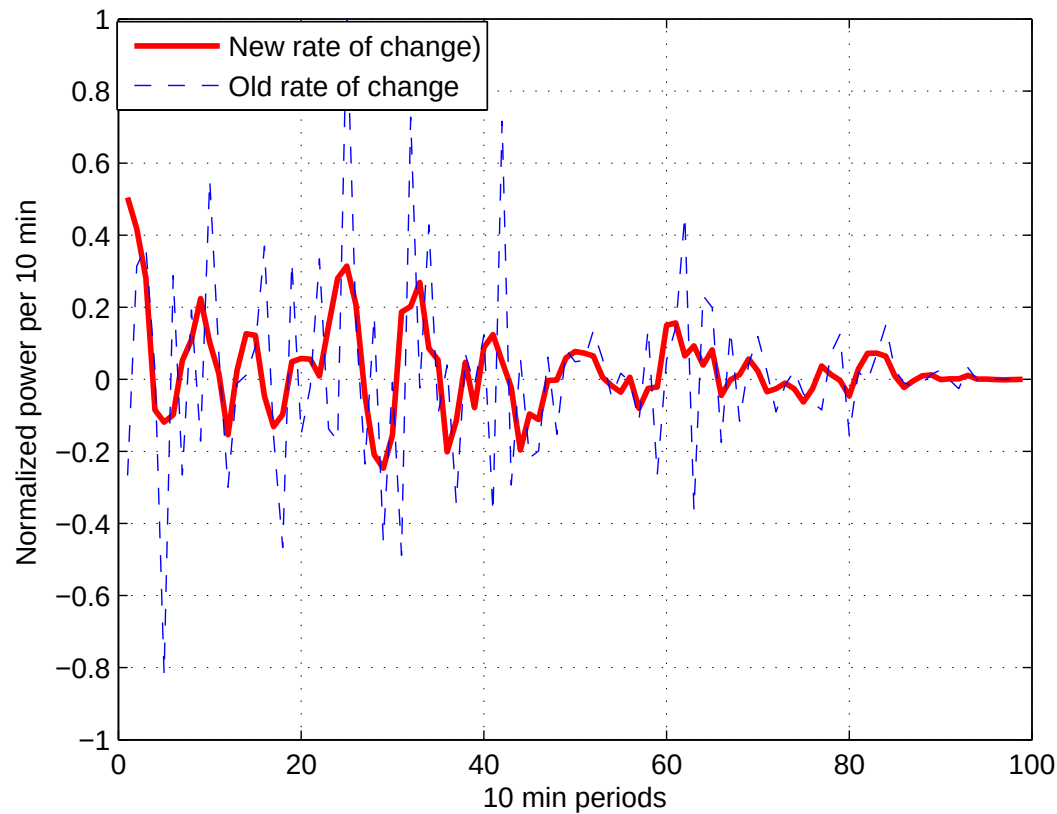
The Control Results

- Normalized actual and smoothed power with individual turbine data.



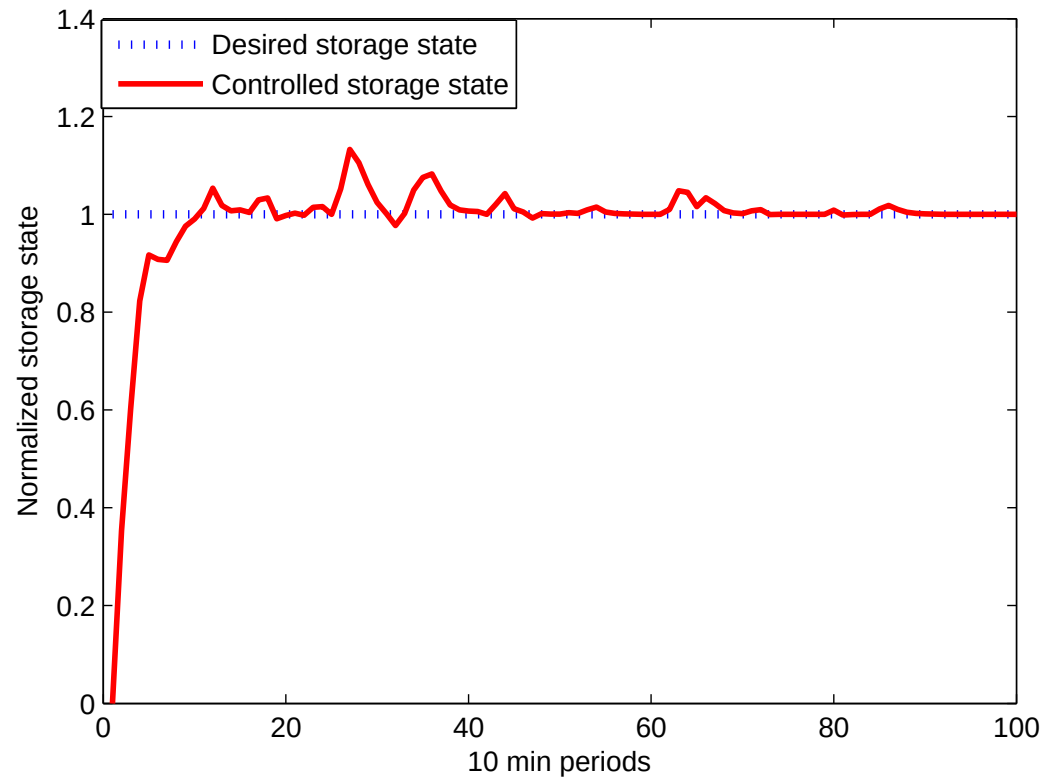
Results (Contd...)

- Ramp rate with individual turbine data.



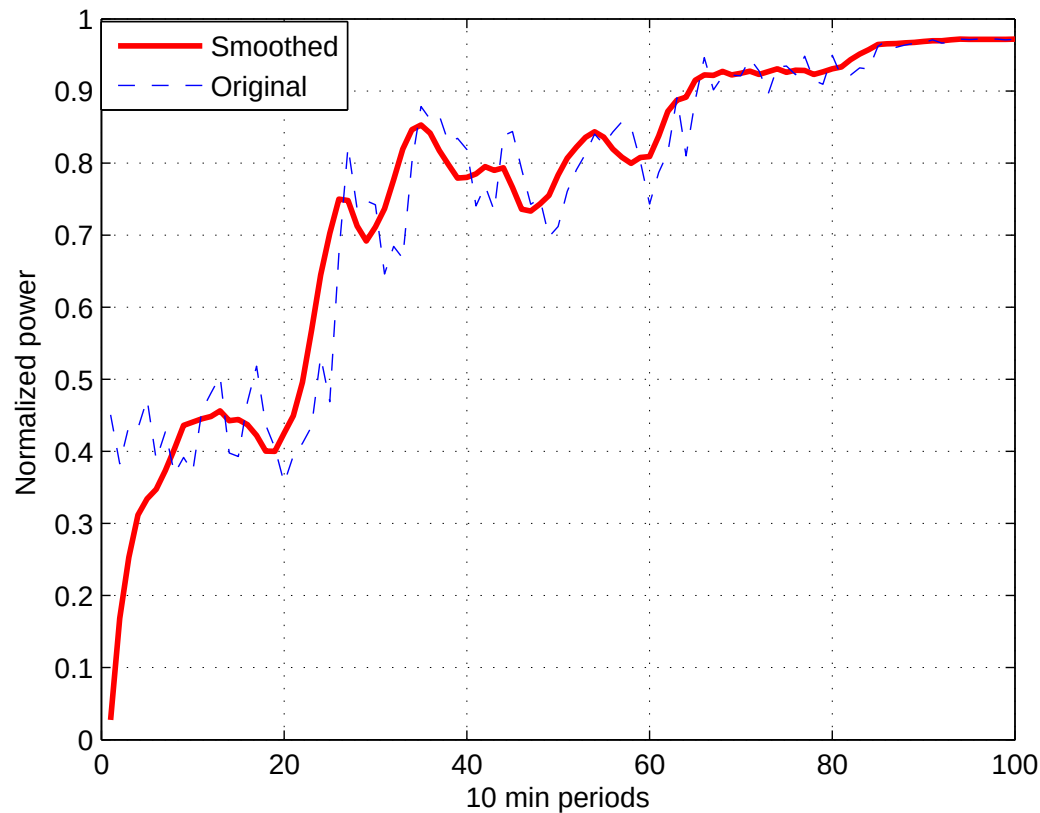
Results (Contd...)

- Storage state with individual turbine data.



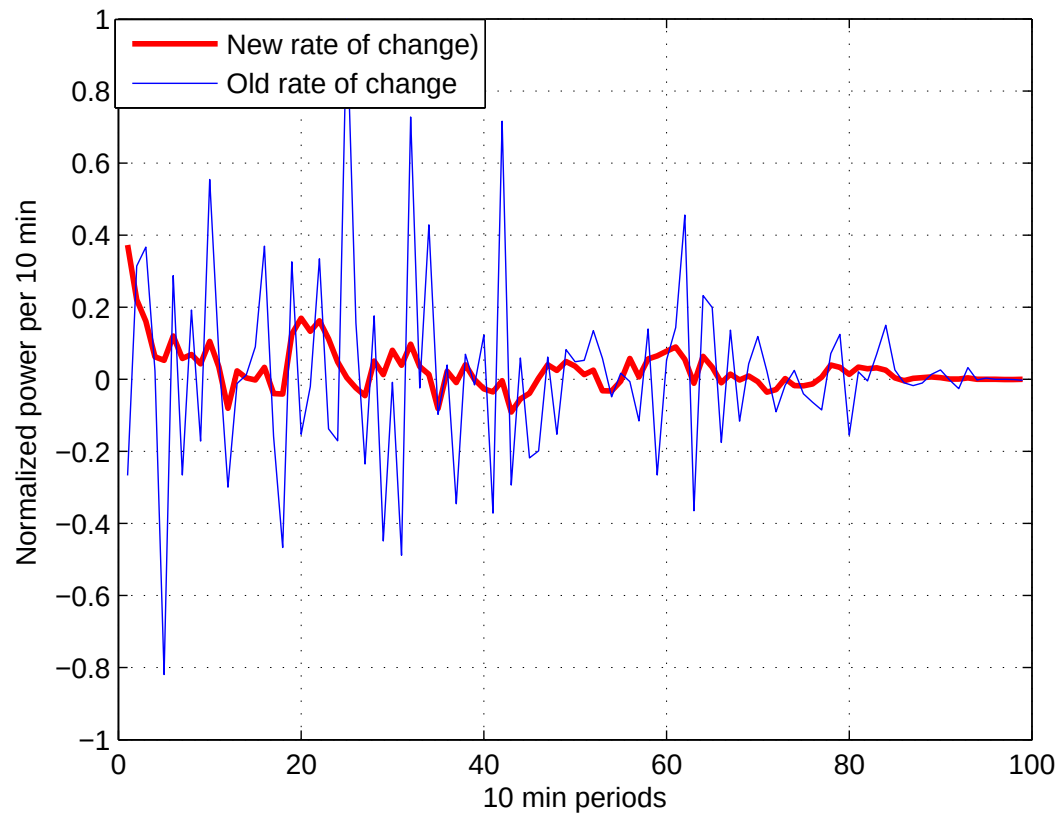
Results (Contd...)

- Normalized actual and smoothed power with averaged turbine data.



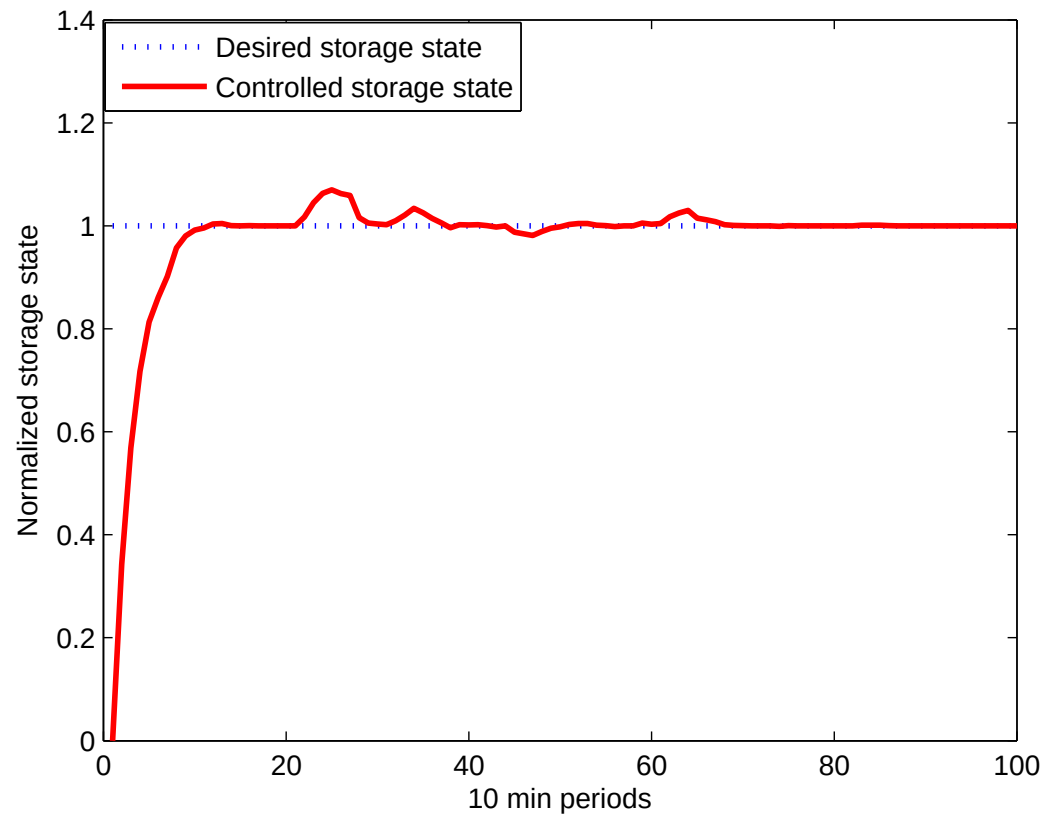
Results (Contd...)

- Ramp rate with averaged turbine data.



The Control Results

- Storage state with averaged turbine data.



Concluding Remarks

- A [Model Predictive Control](#) based controller has been proposed to smooth the highly variable wind power and to optimize the wind energy storage in batteries.
- The proposed controller is not only suitable for wind energy but equally applicable to any intermittent energy source.
- The controller proved to be always stable.
- A [Wind Power Prediction Model](#) has also been proposed to improve the overall control system performance.

Future Research Aims

- The testing of the algorithm with high sampling data.
- The analysis of noise effect on the controller performance.